

Intro to ML — Milestone 1 Report

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1 Introduction

Predicting addictive behaviors from behavioral data is a canonical supervised learning task. We apply five classical methods—Linear Regression, Logistic Regression, k -Nearest Neighbors, Linear SVM, and Kernel Ridge (with its classifier variant)—to the Gaming and Mental Health dataset (2000 synthetic samples, 13 gaming-habit features), across two tasks: regression of a continuous addiction score, and 3-class classification of addiction level (Low, Medium, High). Two properties of the data shape our choices: features span an order of magnitude in scale, and the classification labels are strongly imbalanced, making accuracy a misleading headline. We report MSE for regression, and accuracy with macro-F1 (primary) for classification; hyperparameters are chosen by 5-fold cross-validation.

2 Methodology

2.1 Data & Preprocessing

The 13 raw features span over an order of magnitude in standard deviation (largest std ≈ 514 , smallest ≈ 0.19), so we z-score normalize using training-set statistics. Classification labels are severely imbalanced—Low 68.9%, Medium 27.9%, High 3.2%—motivating stratified sampling and macro-F1 over accuracy: a majority-class predictor reaches 70.0% accuracy but only 0.275 macro-F1. The regression target is right-skewed (mean 2.79, median 2, range 0–10). We use an 80/20 train/test split.

2.2 Cross Validation

Hyperparameters are selected by 5-fold CV on the training portion, with the final model refit on all of it before evaluation. Classification folds are stratified to preserve the 21 $\times$ imbalance ($n=51$ High samples in train would otherwise risk empty validation folds). For Linear Regression we additionally cross-check 5-fold against exact leave-one-out CV via the hat-matrix shortcut $\text{LOO}_i = r_i / (1 - H_{ii})$ (no N refits); the two agree within ± 0.005 MSE across a seven-point λ sweep, while a single held-out split is visibly noisier (~ 0.07 MSE offset). Large search spaces (LogReg’s $\eta \times \beta \times T$; KNN’s $k \times \text{metric} \times \text{weighted}$) use coarse-to-fine refinement.

3 Methods & Results

3.1 Linear Regression

We implemented ridge regression in closed form, $\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$ (bias unregularized), and via gradient descent on the same objective $\|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 + \lambda \|\mathbf{w}\|_2^2$; the two solvers converge to identical weights ($\|\Delta \mathbf{w}\| / \|\mathbf{w}\| \approx 10^{-16}$) but CF is $\sim 500\times$ faster. Polynomial feature expansion (per-feature powers or full cross-terms up to degree d) is also supported.

5-fold CV — validated against exact LOOCV via the hat-matrix shortcut $\text{LOO}_i = r_i / (1 - H_{ii})$, with agreement within ± 0.005 MSE — is flat at ≈ 1.006 for $\lambda \leq 10$ and rises at $\lambda=100$ as over-regularization dominates. Polynomial expansion either ties or hurts: cross-terms ($d=2+\text{int}$) overfit (train 0.92, CV 1.08); $d=3$ without interactions lies within the 0.068 fold std of baseline (Fig. 1). We select the simplest model at the performance floor: closed-form ridge, $\lambda=10$, $d=1$, with test MSE **0.993** (Table 1).

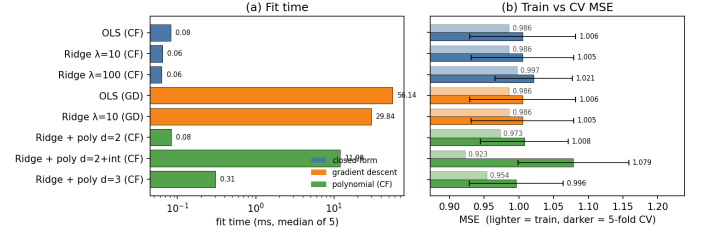


Figure 1: All Linear Regression variants. **(a)** Median fit time (log-scale): closed-form (CF) is $\sim 500\times$ faster than GD; polynomial cross-terms ($d=2+\text{int}$) inflate D from 13 to ~ 100 and slow CF visibly. **(b)** Train (light) vs 5-fold CV MSE (dark, $\pm$ std). All linear solvers tie at the noise floor; $\lambda=100$ under-fits; $d=2+\text{int}$ overfits (largest train–CV gap); $d=3$ matches baseline within CV noise. Selected: CF ridge, $\lambda=10$, $d=1$.

3.2 Logistic Regression

We implemented multinomial softmax logistic regression trained by gradient descent, with options for Xavier init, heavy-ball momentum, early stopping, L2 regularization (bias excluded), inverse-frequency class-weighted cross-entropy, polynomial feature expansion, mini-batch SGD, and inverse-time learning-rate decay.

At a sufficient compute budget ($\eta=0.3$, 1000 iters), every optimizer enhancement converges to macro-F1 ≈ 0.758 in 5-fold stratified CV (Fig. 2b): momentum, Xavier init, early stopping and lr-decay are redundant once the optimizer has enough steps. Mini-batch SGD at $bs=32$ reaches the same F1 at $\sim 4\times$ less fit time (30 ms vs 130 ms); pure SGD ($bs=1$) is too noisy (F1 0.655). The single change that improves the primary metric is *inverse-frequency class weighting*: macro-F1 rises to 0.769 at a 1.4pp accuracy cost—a trade the 21 $\times$ imbalance makes worth taking. Over-regularization collapses minority F1 ($\lambda=100$: F1 0.513), and polynomial cross-terms overfit ($d=2+\text{int}$: F1 0.726). Our final model uses class-weighted CE with momentum, achieving **80.4%** test accuracy and macro-F1 **0.706** on the held-out 20% (Table 1).

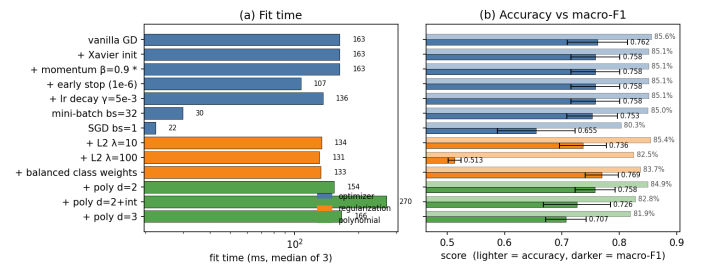


Figure 2: All Logistic Regression variants. **(a)** Median fit time (log scale): mini-batch SGD at $bs=32$ matches full-batch F1 at $\sim 4\times$ less wall-clock; polynomial cross-terms slow training as D grows. **(b)** 5-fold stratified CV accuracy (lighter) and macro-F1 (darker, $\pm$ std). Class weighting is the only knob that raises F1; $\lambda=100$ collapses minority F1 without hurting accuracy as much. Selected: class-weighted CE with momentum.

3.3 k -Nearest Neighbors (KNN)

We implemented a unified KNN for both tasks: majority vote for classification and mean target for regression, with L1, L2 or cosine distance and optional inverse-distance weighting of the neighbors.

A full grid over $k \times \text{metric} \times \text{weighted}$ (5-fold stratified CV for classification, random 5-fold for regression) picks the same winning configuration for both tasks: $k=17$, cosine,

inverse-distance weighted (Fig. 3). Cosine dominates the other two metrics by a clear margin on both panels—after z-score normalization the features already live at comparable scales, so directional (cosine) similarity carries more signal than absolute L1/L2 distance. Small k overfits (classification F1 drops to ≈ 0.58 at $k=1$); large k under-fits; the curves bottom out in the $k \in [10, 20]$ range. Inverse-distance weighting is consistently ~ 0.02 F1 / MSE better than uniform (not shown, for legibility). Predict time on the full $N=1600$ training set is 37 ms (cosine), 58 ms (L2), 238 ms (L1), i.e. KNN’s per-query cost scales linearly in N —its main deployment-time weakness. Selected model ($k=17$, cosine, weighted) achieves 80.1% test accuracy and macro-F1 **0.623** for classification, and test MSE **1.62** for regression (Table 1).

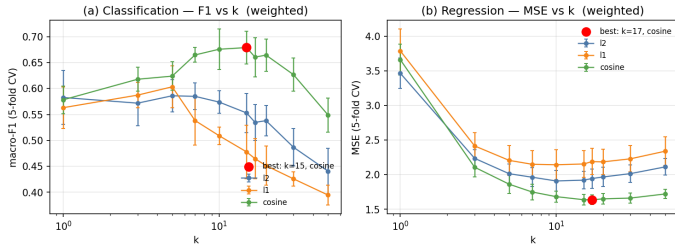


Figure 3: KNN across k with the three distance metrics (inverse-distance weighted; uniform weighting is consistently ~ 0.02 worse and omitted for legibility). (a) 5-fold stratified CV macro-F1 for classification: small k overfits, large k under-fits, the optimum lies at $k=15-20$ for every metric. (b) 5-fold CV MSE for regression: same curve shape, same cosine dominance. Selected: $k=17$, cosine, weighted.

3.4 Linear SVM

We implemented a linear multi-class SVM trained by sub-gradient descent on the Crammer–Singer hinge loss

$$L_i(\mathbf{W}) = \max\left(0, 1 + \max_{c \neq y_i} \mathbf{w}_c^\top \mathbf{x}_i - \mathbf{w}_{y_i}^\top \mathbf{x}_i\right),$$

sharing the kwargs of LogReg (L2, momentum, LR decay, mini-batch SGD, class weighting) so the two are HP-comparable. Since SVM and LogReg share the same linear hypothesis class but optimize different surrogates—margin (hinge) vs probability (log-loss)—a head-to-head comparison isolates the effect of the *loss function* on this dataset (Fig. 4b).

5-fold stratified CV shows the SVM objective is essentially flat across $\lambda \in [10^{-4}, 10^{-1}]$ (macro-F1 ≈ 0.76), with the best point at $\lambda=10^{-2}$ (F1 0.764 ± 0.048). Full-batch sub-gradient descent is $7\times$ slower than mini-batch (336 ms vs 45 ms) for no F1 gain. Unlike LogReg, inverse-frequency class weighting *hurts* SVM F1 (0.746): the hinge already ignores easy-correct points, so up-weighting the minority double-counts examples the model already classifies well. At the shared mini-batch regime, SVM slightly beats LogReg (F1 0.763 vs 0.753; accuracy 85.8% vs 85.0%), suggesting the margin objective handles this near-separable structure marginally better. Selected model (SVM $\lambda=10^{-2}$, bs=64, $\beta=0.9$) achieves **82.3%** test accuracy and macro-F1 **0.713** (Table 1).

3.5 Kernel Ridge & Kernel Least-Squares

Following Lecture 7 we implement (i) Kernel Ridge Regression $\alpha = (\mathbf{K} + \lambda \mathbf{I})^{-1}(\mathbf{y} - \bar{\mathbf{y}})$ and (ii) the multi-output Kernel Least-Squares Classifier $\mathbf{A} = (\mathbf{K} + \lambda \mathbf{I})^{-1} \mathbf{Y}$ on one-hot $\mathbf{Y}$.

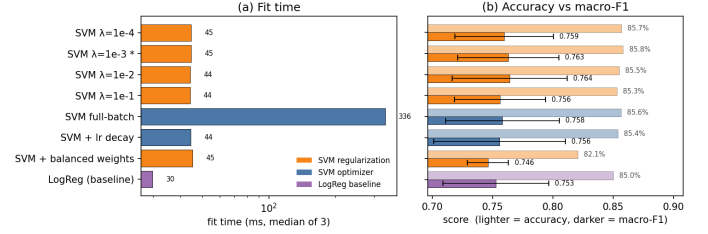


Figure 4: Linear SVM variants, with a LogReg baseline (purple) for head-to-head comparison. (a) Median fit time (log scale): mini-batch $\sim 7\times$ faster than full-batch; both optimizers reach the same F1. (b) 5-fold CV accuracy (lighter) and macro-F1 (darker, $\pm$ std). The SVM objective is flat across $\lambda \in [10^{-4}, 10^{-1}]$; balanced weights hurt (SVM-specific asymmetry with LogReg). At matched settings SVM edges LogReg by ~ 0.01 F1.

Three kernels: linear $\mathbf{x}^\top \mathbf{y}$, polynomial $(\mathbf{x}^\top \mathbf{y} + c)^d$, and RBF $\exp(-\|\mathbf{x} - \mathbf{y}\|^2 / 2\sigma^2)$. We centre $\mathbf{y}$ before solving KRR to recover the intercept the lecture form omits; KLS’s argmax is shift-invariant so no centering is needed there.

For regression, the *linear* kernel ties LinReg at train MSE to machine precision ($\|\Delta\| \approx 10^{-16}$), confirming the implementation; RBF needs $\sigma \geq 10$ to match it (a wide RBF locally approximates linear), and polynomial $d=3$ overfits badly (CV MSE 2.61, Fig. 5a). For classification, the lecture-form *linear* kernel underperforms (KLS F1 0.528) because the bias-free dual cannot represent per-class offsets; polynomial $d=2$ and RBF $\sigma=3$ recover $F1 \approx 0.72$ but still trail LogReg’s class-weighted 0.769 (Fig. 5b). Final models: **KRR linear** (test MSE **0.969**) and **KLS poly $d=2$** (test acc 81.7%, F1 **0.619**) (Table 1).

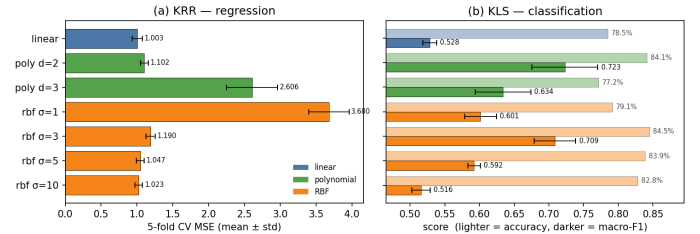


Figure 5: Kernel sweep at $\lambda=1$, 5-fold CV. (a) KRR regression MSE: linear ties LinReg (MSE 1.003); RBF approaches it only for wide σ ; polynomial $d=3$ overfits. (b) KLS classification accuracy (light) and macro-F1 (dark, $\pm$ std): polynomial $d=2$ wins; the bias-free linear kernel underperforms.

4 Discussion

4.1 Latency and scalability

Fit and predict costs split the methods sharply. Closed-form LinReg fits in ~ 0.1 ms ($\sim 500\times$ faster than GD). LogReg and SVM mini-batch take 30–60 ms. KNN’s predict is $O(N)$ per query (37–238 ms for $N=1600$, metric-dependent), scaling linearly in training-set size—ruling it out for latency-sensitive deployment. Kernel methods sit at the opposite extreme: the $N \times N$ solve is $O(N^3)$ in fit and $O(N)$ in predict, with $O(N^2)$ memory. For this dataset ($N=1600$) all methods run comfortably, but the scaling gap would dominate at $N \gtrsim 10^4$: only the parametric linear models (LinReg, LogReg, SVM) would remain practical at that scale.

4.2 Best models per task

Table 1 consolidates the selected configuration and held-out test performance for every method. **Regression** is essentially a linear problem: KRR with the linear kernel (test MSE 0.969) narrowly edges ridge LinReg (0.993), both within fold-to-fold noise, while KNN trails at 1.62. **Classification** is tighter: SVM ($\lambda=10^{-2}$) achieves the best test macro-F1 (0.713), a modest ~ 0.01 margin over class-weighted LogReg (0.706); KLS (0.619) and KNN (0.623) trail by ~ 0.1 F1, driven by poor recall on the 11-sample High test minority. The margin-vs-log-loss difference matters slightly on this dataset, but both parametric linear classifiers beat the non-parametric and kernel alternatives.

Table 1: Selected configurations and test-set performance. Reg: MSE (lower = better); Clf: accuracy, macro-F1 (higher = better).

Method	Task	Config	Test
LinReg	reg	CF ridge $\lambda=10$	MSE 0.993
KRR	reg	linear kernel, $\lambda=1$	MSE 0.969
KNN	reg	$k=17$, cos, wtd	MSE 1.62
LogReg	clf	bal. weights, $\eta=0.3$, $\beta=0.9$	80.4% / F1 0.706
SVM	clf	$\lambda=10^{-2}$, bs= 64	82.3% / F1 0.713
KLS	clf	poly $d=2$, $\lambda=1$	81.7% / F1 0.619
KNN	clf	$k=17$, cos, wtd	80.1% / F1 0.623

5 Conclusion

Five classical methods on the Gaming and Mental Health dataset converge on one story: the signal is near-linear in the 13 z-scored features, so parametric linear models win both tasks by small margins. Regression goes to KRR with a linear kernel (test MSE 0.969); classification to Linear SVM (test F1 0.713). The remaining bottleneck is minority-class recall on the 3.2% High class—a data problem more than a model problem, and a natural target for future work.